

Amine-Functionalized Boron Nitride Nanosheets: A New Functional Additive for Robust, Flexible Ion Gel Electrolyte with High Lithium-Ion Transference Number

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Ion gel electrolytes show great potential in solid-state batteries attributed to their outstanding characteristics. However, because of the strong ionic nature of ionic liquids, ion gel electrolytes generally exhibit low lithium-ion transference number, limiting its practical application. Amine-functionalized boron nitride (BN) nanosheets (AFBNNs) are used as an additive into ion gel electrolytes for improving their ion transport properties. The AFBNNs-ion gel shows much improved mechanical strength and thermal stability. The lithium-ion transference number is increased from 0.12 to 0.23 due to AFBNNs addition. More importantly, for the first time, nuclear magnetic resonance analysis reveals that the amine groups on the BN nanosheets have strong interaction with the bis(trifluoromethanesulfonyl)imide anions, which significantly reduces the anion mobility and consequently increases lithium-ion mobility. Battery cells using the optimized AFBNNs-ion gel electrolyte exhibit stable lithium deposition and excellent electrochemical performance. A $\text{LiFePO}_4/\text{Li}$ cell retains 92.2% of its initial specific capacity after the 60th cycle while the cell without AFBNNs-gel electrolyte only retains 53.5%. The results not only demonstrate a new strategy to improve lithium-ion transference number in ionic liquid electrolytes, but also open up a potential avenue to achieve solid-state lithium metal batteries with improved performance.

and high safety levels. Lithium metal batteries (LMBs), owing to the low redox potential (-3.04 V vs standard hydrogen electrode) and ultrahigh theoretical capacity (3860 mAh g^{-1}) of the lithium (Li) metal anode, are promising to fulfill these requirements.^[1] However, the organic liquid electrolyte (LE), currently widely used in the LMBs, makes LMBs face severe safety concerns such as electrolyte leakage and combustion due to the intrinsic fluidity, flammability, and electrochemical instability. Replacing the organic liquid electrolyte with solid-state electrolytes (SSEs) has been proven to be an efficient way to overcome these problems.^[2] Because of their high mechanical stability and structural integrity, SSEs enable facile cell fabrication and prevent electrolyte leakage, offering greater safety than liquid electrolytes. Current solid-state electrolytes include solid polymer electrolytes (SPEs) and inorganic electrolytes (IEs). Unfortunately, SPEs suffer from a poor ionic conductivity at room temperature, while the IEs show innate brittleness and intrinsically narrow electrochemical stability along with high contact resistance. Compared to

SPEs and IEs, gel electrolytes, which are prepared by entrapping liquid electrolytes into a solid matrix, provide exceptional properties such as high ionic conductivity, low interfacial resistance, good mechanical strength, and flexibility.

Liquid mediums are one of the key components in gel electrolytes. Among the candidates, ionic liquids (ILs), also known as room temperature molten salts, can result in high performance gel electrolytes because of their unique characteristics of chemical, electrochemical, and thermal stability, negligible volatility, and nonflammability.^[3] Moreover, IL-based electrolytes (ILEs) can assist the stabilization of Li anodes through the formation of a robust solid electrolyte interphase (SEI) during cycling, and consequently enhance cycle life and render better safety.^[4] When these IL-based electrolytes are immobilized into a solid matrix, as-prepared ion gel electrolytes can act as sustainable SSEs for all-solid-state batteries.^[5] Nevertheless, ILE generally exhibit very low lithium-ion transference numbers (t_{Li^+}) due to the strongly ionic nature of ILs as a solvent, which

1. Introduction

The rapid development of electric vehicles and flexible/wearable electric devices spurs the relentless pursuit of advanced energy storage devices with high energy density, long cycle life,

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strongly limits the application of ILEs. It has been reported that electrolytes with high t_{Li^+} can significantly inhibit the growth of Li dendrites because of the suppression of space charge formation at the surface of Li anodes.^[6] Therefore, ILEs with high t_{Li^+} are urgently desired and thus various strategies have been proposed to increase the t_{Li^+} of ILEs. Among them, developing functional additives is the most promising strategy to improve the mobility of lithium ions in the SSE, and examples include SiO_2 ,^[7] Al_2O_3 ,^[8] and aluminosilicate clays.^[9] However, how the additives improve the lithium-ion conductivity and t_{Li^+} has not been revealed clearly, and the low t_{Li^+} of the ILEs is still unsatisfactory. Therefore, new functional additives are highly desired, and the mechanism needs to be clarified.

Boron nitride nanosheets (BNNSs), as an analogue of graphene, exhibit special characteristics of high thermal conductivity, low electrical conductivity, chemical inertness, and high mechanical strength.^[10] Moreover, BNNS can be functionalized with different functional groups for specific applications.^[11] Here, an ion gel electrolyte with amine-functionalized boron nitride nanosheets (AFBNNSs) is designed and fabricated via a thermal polymerization process. The obtained ion gel electrolyte demonstrates an excellent mechanical strength of nearly two times higher than the strength of the ion gel without AFBNNs. More importantly, the obtained ion gel electrolyte possesses a high lithium

transference number of 0.23, which is almost a twofold enhancement compared with the ion gel electrolyte without AFBNNs addition. The improvement mechanism of the AFBNNs has been investigated in detail using both spectroscopic measurements and nuclear magnetic resonance (NMR) analysis. It is found that the amine groups on the BN nanosheets have a strong interaction with the bis(trifluoromethanesulfonyl)imide (TFSI^-) anions which significantly reduces the mobility of the TFSI^- , leading to an increased lithium-ion mobility and lithium transference number of the ion gel electrolyte. Due to the excellent mechanical properties and high ion transport, the ion gel electrolyte with AFBNNs addition shows advantages for suppressing Li dendrites demonstrated by finite-element simulation results and measured long and stable cycles of symmetric Li cells. Furthermore, several lithium metal batteries, including LiFePO_4 (LFP)|Li, LiCoO_2 (LCO)|Li, and $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ (NCM)|Li, demonstrate a higher capacity and longer cycling life with the AFBNNs electrolytes.

2. Results and Discussions

Figure 1a illustrates the synthesis process for the AFBNNs-added gel polymer electrolyte (BN GPE). A pyrrolidinium cation-based IL is used as a solvent for preparation of the ILE medium,

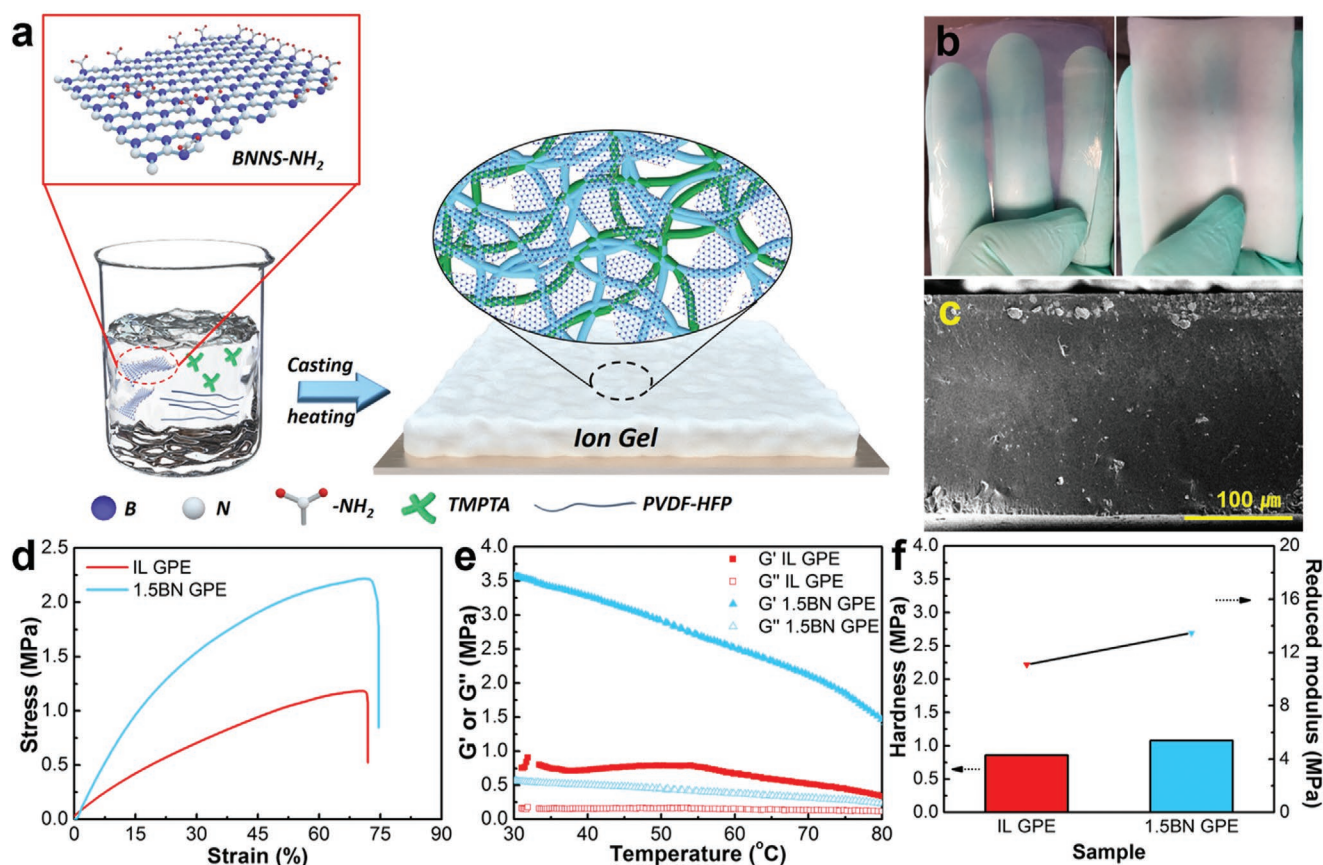


Figure 1. a) Schematic diagram of preparation process of functionalized BN nanosheet added ion gel. b) Optical photograph of the as-prepared ion gel holding by a hand. c) SEM images of ion gel film (cross-section) with functionalized BN nanosheet addition. d) Tensile stress–strain curve, e) storage modulus (G') and loss modulus (G'') versus temperature, and f) hardness and reduced modulus of IL GPE and 1.5BN GPE.

because of its wider electrochemical stability window than imidazolium cation-based ILs and its favorable effect on SEI formation on the Li metal anode.^[4] With this ILE, we prepared the ion gel electrolyte through in situ thermal polymerization of trimethylolpropane triacrylate (TMPTA) with poly(vinylidene fluoride-co-hexafluoropropylene) (PVDF-HFP). TMPTA with three double bonds acting as crosslinker offers high mechanical strength through the construction of a crosslinking network on the robust and flexible PVDF-HFP chains, similar to the case reported by Xia et al.^[12] Additionally, the carbonate functionality of TMPTA will contribute to enhanced Li⁺ transfer in the prepared ion gel.^[13] BN nanosheets were functionalized with amine groups (Figure S1, Supporting Information) to act as efficient additives. The polymerization of the ion gel was confirmed by the absence of an acrylic C=C bond peak around 1630 cm⁻¹ in the Fourier-transform infrared spectroscopy (FT-IR) spectra (Figure S2, Supporting Information). The photographs of as-prepared ion gels (Figure 1b) show self-standing ability and good flexibility. Figure 1c shows the cross-sectional scanning electron microscope (SEM) image of the prepared BN GPE film. Different from cold-pressed inorganic electrolytes or solvent-cast solid polymer electrolytes, the obtained GPE displays solid morphology without voids, which will accelerate the Li⁺ transport because of the elimination of the transport barrier.^[2b,14] The thickness of BN GPE is around 205 ± 5 μm, and the addition of AFBNNs into the ion gel merely affects the film thickness (Figure S3, Supporting Information). Energy-dispersive X-ray (EDX) elemental mapping further shows a uniform distribution of BNNs into the ion gel matrix without notable aggregation (Figure S4, Supporting Information).

Mechanical properties were investigated with uniaxial tensile measurement (UTM) (Figure 1d). BN GPE exhibits a maximum strength of 2.22 MPa with a maximum strain rate of 72%, which is nearly two times higher than the strength of the starting IL GPE (1.18 MPa). The results indicate the mechanical strength of ion gel can be much improved with the addition of the AFBNNs. Dynamic mechanical analysis (DMA) tests further reveal that both ion gels show a storage modulus (*G'*) higher than the loss modulus (*G''*), indicating solid-like behavior. Notably, *G'* of 1.5BN GPE is far higher than that of the IL GPE. Mechanical stability at the surface was also evaluated by indentation measurement (Figure 1f). Again, both hardness and reduced modulus of ion gels are improved with the addition of AFBNNs, which will contribute to mitigating of Li dendrite formation during the charge–discharge cycling of the battery.^[1b,15] All these results demonstrate that the AFBNNs substantially improve the mechanical properties of ion gels through interfacial interactions.^[16] In addition, the crystalline nature of the prepared ion gels was further investigated, as the crystallinity of the polymer matrix can significantly affect the ion transport, i.e., most poly(ethylene oxide) (PEO)-based polymer electrolytes suffer from recrystallization of PEO chains.^[17] The X-ray diffraction (XRD) patterns show that the characteristic peaks of α -phase polyvinylidene fluoride (PVDF) disappear after gelation, which is attributed to the plasticizing effect of the IL (Figure S5a, Supporting Information).^[18] The differential scanning calorimetry (DSC) curve (Figure S5b, Supporting Information) shows the melting temperature (*T_m*) of PVDF-HFP at around 146 °C. The gelation of ILE into the

PVDF-HFP matrix decreases *T_m* to 119 °C. After the addition of AFBNNs into IL GPE, *T_m* further decreases to 115 °C, therefore AFBNNs addition reduces the degree of crystallinity and generates more amorphous phase, which is beneficial to fast ionic conduction.^[19]

The thermal stability of the obtained electrolytes, which determines the safety of batteries, was also investigated by thermogravimetric analysis (TGA). The TGA curves show that ether-based LE readily decomposes even at room temperature (Figure S6, Supporting Information). On the other hand, all the ion gel electrolytes display no notable degradation until 330 °C, demonstrating high thermal stability of ion gel electrolytes. The flammability of the as-prepared ion gel electrolytes was further evaluated by directly heating with an ignition source. When a commercial Celgard 2400 separator was heated with flame, rapid shrinkage within 1 s was observed and it was completely deformed in 3 s. In contrast, under the same burning condition, both IL GPE and 1.5BN GPE retained their shape without significant deformation over 12 s, which is attributed to the high thermal stability of the IL and AFBNNs addition.

The ionic conductivities of the electrolytes were measured using electrochemical impedance spectroscopy (EIS) with stainless steel symmetric cells. An organic liquid electrolyte, 1 M LiTFSI in dioxolane (DOL)/dimethoxyethane (DME) solvent (1:1 in volume) was also measured and henceforth denoted as LE. Figure 2a shows the temperature dependencies of ionic conductivity for the investigated electrolytes. The activation energies of the electrolytes were estimated using the Arrhenius equation

$$\sigma(T) = A \exp(-E_a/RT) \quad (1)$$

where $\sigma(T)$ is the ionic conductivity at the temperature *T*, *A* is the pre-exponential factor, *E_a* is the activation energy, and *R* is the ideal gas constant, respectively. All the investigated electrolytes show Arrhenius behavior, demonstrating the thermally activated conduction process. The Arrhenius-like conductivity behavior indicates electrolyte with more amorphous characteristics when compared with the highly crystalline systems following Vogel–Tammann–Fulcher (VTF) relationships.^[20] The calculated *E_a* values suggest that IL GPE possess a lower activation energy of 22.22 kJ mol⁻¹ than that of ILE (25.52 kJ mol⁻¹), while the activation energy of IL GPE is further decreased to 20.14 kJ mol⁻¹ with the AFBNNs addition. Figure 2b shows the ionic conductivities of different samples. The ionic conductivity of ILE with a commercial Celgard2400 separator is 0.31 × 10⁻⁴ S cm⁻¹ at 25 °C with the lowest *E_a* of 25.52 kJ mol⁻¹, which is considerably lower when compared with other electrolytes due to the poor wettability and high viscosity of ILE (Figure S7, Supporting Information). The ionic conductivities of both IL GPE and BN GPE are much higher than that of ILE, which can be attributed to the formation of a homogeneous gel between ILE and PVDF-HFP-based polymer matrix. Meanwhile, BN GPE shows a higher ionic conductivity of 6.47 × 10⁻⁴ S cm⁻¹ at 25 °C than that of the IL GPE (3.98 × 10⁻⁴ S cm⁻¹ at 25 °C). These results clearly demonstrate that AFBNNs can significantly increase the ionic conductivity of the IL GPE. In addition, the ionic conductivity of IL GPE with different ratios (0.5, 1, 1.5, 2.0, and 2.5 wt%) of AFBNNs were also measured and

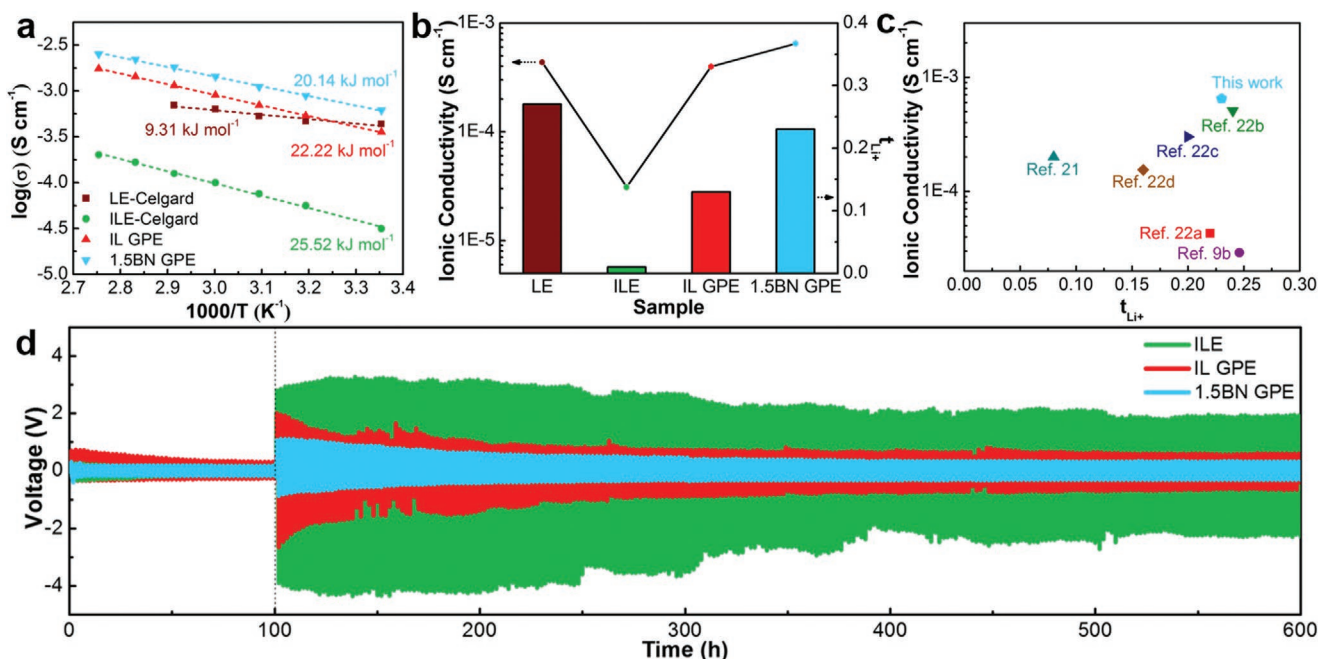


Figure 2. a) The ionic conductivities of investigated electrolytes as a function of temperature. b) t_{Li+} and ionic conductivity of electrolytes at 25 °C. c) Comparison with previously reported electrochemical performances of ion gel electrolytes and other solid-state electrolytes. d) Cycling stabilities of ILE, IL GPE, and 1.5BN GPE in the Li symmetric cell at the current density of 0.1 mA cm⁻² followed by the current density of 0.2 mA cm⁻² with a capacity of 0.2 mAh cm⁻² and 25 °C.

the results are shown in Figure S8a in the Supporting Information. One can see that 1.5BN GPE possesses the highest ionic conductivity. We presume that with increasing the amount of AFBNNs in the ion gel from zero to 1.5 wt%, the interaction between TFSI⁻ and AFBNNs is increased, thereby the ionic conductivity of ion gel is enhanced. However, too much AFBNNs (>1.5 wt%) in the ion gel might block the conduction pathway, which leads to the decrement of ionic conductivity, similar as previously reported graphene oxide added ion gels.^[19] When functionalized BN nanoparticles were added into ion gel, an ionic conductivity enhancement was not observed possibly because of the smaller surface of BN nanoparticles that cannot offer enough sites to immobilize TFSI⁻ anions (Figure S8b, Supporting Information). On account of the high electrochemical stability of the IL, both IL GPE and 1.5BN GPE exhibit a wide potential window over 4.5 V (Figure S9, Supporting Information). The t_{Li+} of investigated electrolytes was obtained by sequential measurement of potentiostatic polarization and impedance spectroscopy (Figure S10, Supporting Information). Because of the reduction in anion mobility by AFBNNs, 1.5BN GPE shows enhanced t_{Li+} of 0.23, which is comparable to ether-based LE t_{Li+} of 0.27 and higher than other reported ion gels using bulk BN of 0.08^[21] or size-controlled BN of 0.18.^[2e] The obtained values are better or comparable to the recently reported values (Figure 2c).^[22] The t_{Li+} of ILE exhibits only 0.02, which implies the strong ionic nature of IL hinders dissociation of Li salts.^[23]

The Li deposition behavior of the BN GPE was investigated in detail. A galvanostatic cycling test was performed by repeatedly charging and discharging for 1 h in a Li symmetric cell at the current density of 0.1 mA cm⁻² and 25 °C (Figure S11,

Supporting Information). Although ion gel electrolytes display slight overpotential at the initial cycle, IL GPE and 1.5BN GPE show stable cycling behavior over 1000 cycles afterward. The good cycling stability of ion gels and ILE is attributed to the favorable effect of the PYR₁₃⁺ cation on SEI formation.^[4] Previous reports showed that IL-based electrolytes could exhibit good cycling stability at a current density of 0.1 mA cm⁻², but few reports successfully demonstrated stable deposition behavior over 0.1 mA cm⁻² without the support of organic solvents.^[2e,22b,24] To assess the cyclability of the prepared ion gels at the higher current densities, Li deposition and stripping was conducted at the current density of 0.2 mA cm⁻² (Figure 2d). ILE exhibits serious overpotential of 4 V and continuously fluctuates after the following cycles, which indicates that a lower t_{Li+} associated with the high viscosity of ILE seriously impedes stable Li deposition. Interestingly, after the initial activation process with overpotential about 1.5 V, the 1.5BN GPE-based Li symmetric cell shows stable cycling behavior over 500 h. The IL GPE-based cell displays similar behavior as 1.5BN GPE, but shows higher overpotential than that of 1.5BN GPE. Thus, stable Li plating/stripping performance of 1.5BN GPE over ILE and IL GPE proves that improved t_{Li+} actually induces uniform Li deposition.

The simulation of Li deposition process was carried out with COMSOL to explore the reason why 1.5BN GPE induces more uniform Li deposition than ILE and IL GPE. The Li deposition behavior was simulated on a Li foil with a tip under a current density of 20 mA cm⁻² for 200 s. Figure 3a shows the current density distribution in ILE during Li deposition, where an uneven current density distribution can be observed and the current density concentrates only at the top of the tips. At the

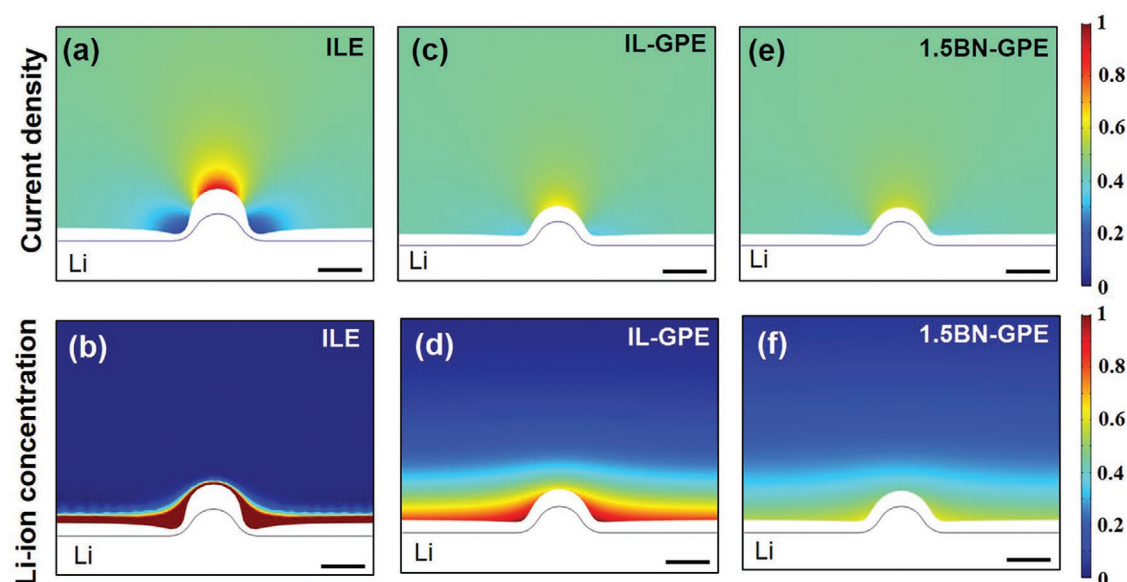


Figure 3. The simulated Li deposition on Li foil with different electrolytes at a simulation time of 200 s. The Li is represented as white space. The normalized local current density profile and Li-ion concentration distribution in the electrolyte during the Li deposition. a,b) ILE, c,d) IL GPE, and e,f) 1.5BN GPE.

same time, the Li ions distribute unevenly in ILE and a large concentration polarization exists (Figure 3b). Because of the concentration polarization of Li, more Li deposits at the top of the tip while less Li deposits at the bottom of the tip, resulting in an uneven Li deposition. When the electrolyte is changed to IL GPE, the current density distributes much more uniformly in IL GPE than that in ILE (Figure 3c). However, there is still a concentration polarization in IL GPE, which is smaller than that in ILE (Figure 3d). Furthermore, when the electrolyte is changed to 1.5BN GPE, both the current density and the Li ions distribute evenly in the 1.5BN GPE (Figure 3e,f), leading to a uniform Li deposition on the electrode. The simulation results

clearly demonstrate that the Li deposition in 1.5BN GPE is more uniform than those in both ILE and IL GPE, which is in accordance with the experimental results. Based on the space-charge theory,^[25] when an electrolyte with a high t_{Li^+} is applied, the space charge formation can be significantly suppressed at the surface of the Li anode, which leads to an even deposition of Li. Therefore, the improvement on the Li deposition behaviors is strongly related to the increased Li ion transference number and Li ion conductivity of the electrolytes.

The improvement of lithium-ion transport in the ion gel is attributed to the reduced mobility of TFSI⁻ counter anions by AFBNNs (Figure 4). With electron-deficient B atoms in their

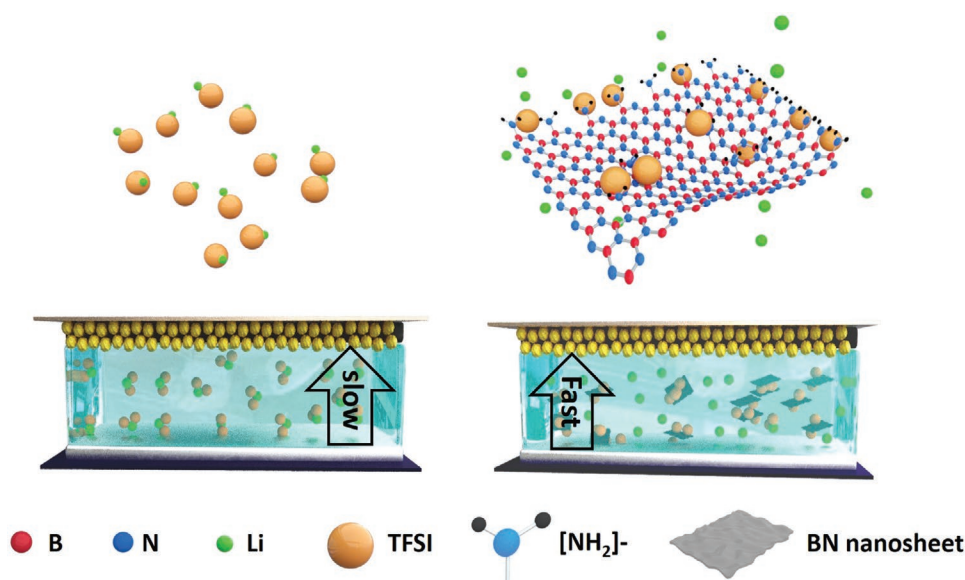


Figure 4. Schematic diagram of Li⁺ ion conduction in the IL GPE and 1.5BN GPE.

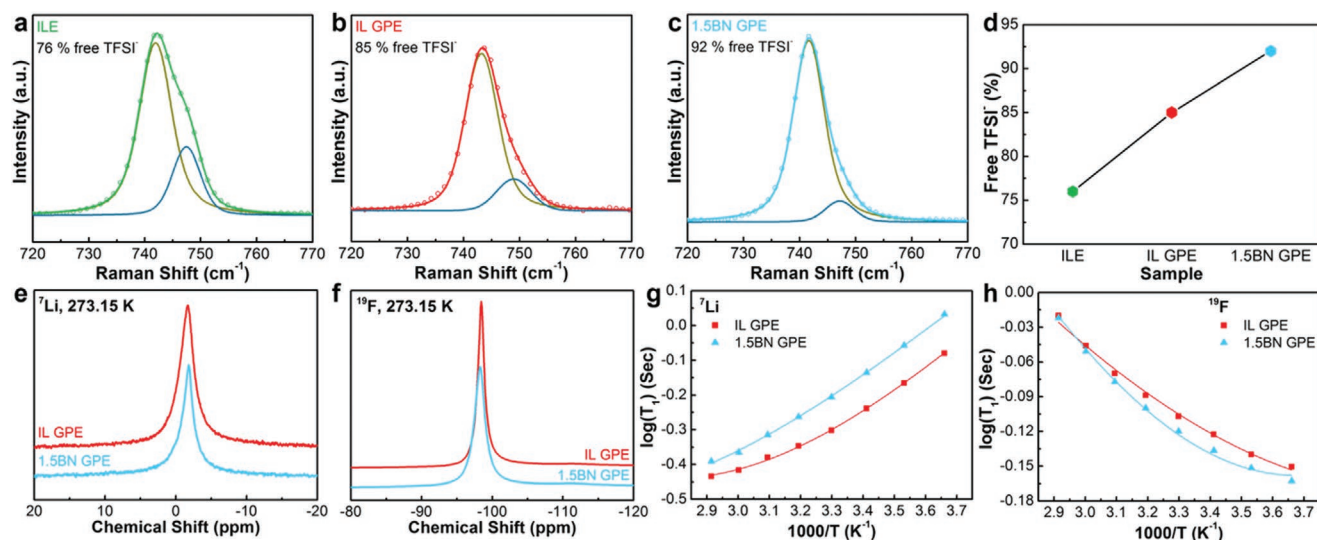


Figure 5. Raman and NMR investigations of ion gel. Deconvoluted Raman spectra of a) ILE, b) IL GPE, and c) 1.5BN GPE, where dot points are raw data obtained. d) Estimated free TFSI⁻ amount in the investigated electrolytes. e) ⁷Li and f) ¹⁹F solid-state NMR spectra. Temperature dependencies of g) ⁷Li and h) ¹⁹F longitudinal T₁ relaxation time.

structure, BNNs exhibit Lewis acidic characteristics that could interact with other Lewis bases.^[26] We further functionalized BNNs with amine-function groups, to further improve the restriction of anion movement, thereby leading to the enhancement of the lithium-ion mobility. To elucidate the improvement mechanism of the ionic conductivity and the lithium transference number of the IL GPE, a series of characterizations including FT-IR, Raman, and NMR were employed. A spectroscopic investigation was first applied to observe an interaction between ILE and BNNs. FT-IR spectrum shows a slight peak shift assigned to TFSI⁻ anion after the addition of BNNs, indicating the presence of interaction between the TFSI⁻ anions and BNNs (Figure S12, Supporting Information). Raman spectra were further collected around the $\approx 745\text{ cm}^{-1}$ region, where this band is an indicator of the Li⁺ coordination state with TFSI⁻ anion (Figure 5).^[27] The Raman peak at around 745 cm^{-1} could be deconvoluted into two components: the high shift of 748 cm^{-1} is associated to Li⁺ coordinated by TFSI⁻ anions while the low shift of 742 cm^{-1} can be assigned as free TFSI⁻ anions not interacting with the Li⁺ cations. While ILE shows a free TFSI⁻ anion intensity of only 76%, IL GPE displays 85% of Li⁺-uncoordinated anions due to the interaction between the anions and the fluorinated polymer chains.^[28] 1.5BN GPE exhibits 92% of Li⁺-uncoordinated anions, demonstrating a larger content of dissociated Li⁺ ions through immobilization of TFSI⁻ anion by the functionalized BNNs addition. These trends are consistent with the obtained t_{Li^+} values of the corresponding electrolytes (Figure 2b).

Although a reduction in activation energy for the ionic conductivity of the ion gel was observed after AFBNNs addition, this cannot directly explain the improvement of Li⁺ ion conduction because the IL as a solvent with ionic characteristics also contributes to ionic transport in the ILE. To probe differences in the mobilities of the Li⁺ and TFSI⁻ ions in these samples, solid-state NMR signals were collected from 0 to 70 °C (Figure 5e,f). ⁷Li NMR spectra show that the peaks from 1.5BN

GPE possess a narrower full width at half maximum (FWHM) than that of IL GPE, i.e., 217 and 281 Hz for 1.5BN GPE and IL GPE, respectively. The narrower ⁷Li NMR peak for 1.5BN GPE is consistent with an increased translational Li⁺ mobility in the ion gel through AFBNNs. In the case of ¹⁹F NMR spectra, this trend is reversed. The ¹⁹F IL GPE NMR spectrum exhibits a narrower peak with FWHM of 310 Hz, while the ¹⁹F 1.5BN GPE NMR spectrum displays a broader peak with FWHM of 509 Hz, consistent with both a suppressed mobility of the TFSI⁻ anions and a greater range of surrounding structural environments due to the interaction with the AFBNNs. Fast timescale, local motions of the ions were also probed by measuring ⁷Li and ¹⁹F T₁ relaxation times as a function of temperature. The results are shown in Figure 5g,h, and fitted using Bloembergen–Purcell–Pound (BPP) theory.^[29] The calculated E_a from ⁷Li T₁ is 13.62 and 15.09 kJ mol⁻¹ for 1.5BN GPE and IL GPE, respectively, again consistent with enhanced mobility of Li⁺ ions in the 1.5BN GPE ion gel. The considerably lower E_a obtained by T₁ measurement than E_a obtained by ionic conductivity measurement is due to the dependence of T₁ on fast timescale, local ionic dynamics (on the order of nanoseconds).^[30] Estimated E_a results from ¹⁹F T₁ further reveal an increased energy barrier for the TFSI⁻ anion in the 1.5BN GPE, i.e., 9.76 kJ mol⁻¹ for 1.5BN GPE and 6.69 kJ mol⁻¹ for IL GPE. This could be explained by the interaction of the AFBNNs with the TFSI⁻ anions, thereby restricting the rotational motions of the anions. This interaction would promote ion dissociation and generate more free Li⁺ cations in the ion gel, consequently enhancing the ionic conductivity and t_{Li^+} of as-prepared ion gel electrolytes.

To evaluate the feasibility of the AFBNNs ion gel in a real-time application, LiFePO₄/Li cells with 1.5BN GPE were assembled. For comparison, cells with ILE and IL GPE were also fabricated and measured. Figure 6a shows the cycling performances of LFP cathode in the voltage window of 2.5–4.2 V. The cells were first activated with 3 cycles at 0.1C, followed by cycling at 0.2C. Compared to LFP with ILE and IL GPE, the

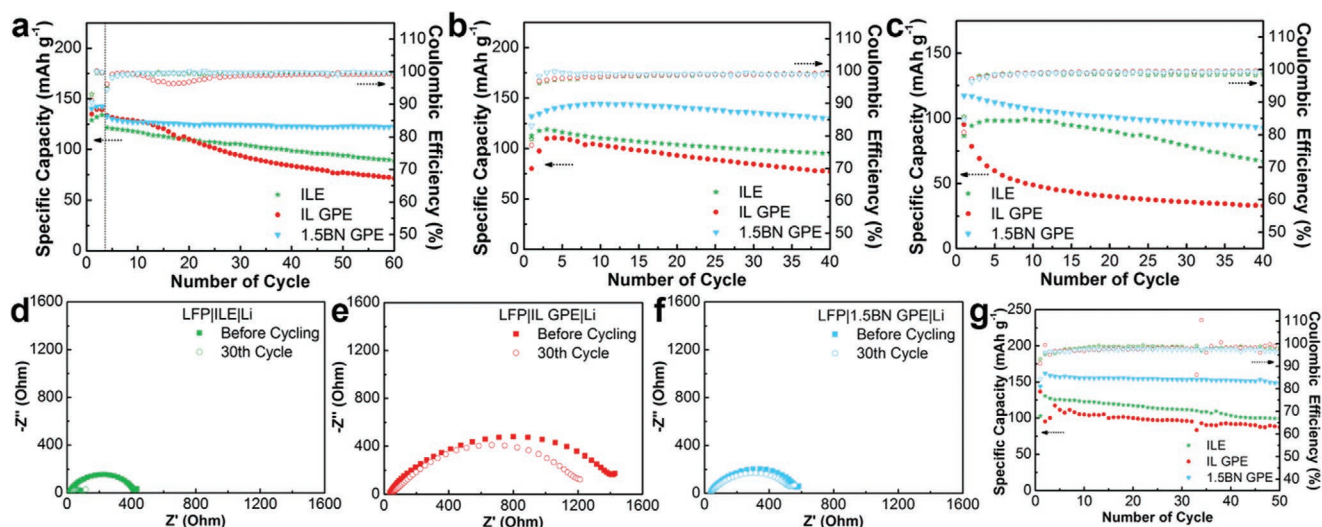


Figure 6. Cycling stability of a) LFP|Li cell, b) NCM|Li cell, and c) LCO|Li cell with ILE, IL GPE, and 1.5BN GPE as the electrolytes. EIS spectra of the d) LFP|ILE|Li, e) LFP|IL GPE|Li, and f) LFP|1.5BN GPE|Li cells at before cycling and after 30th cycle. g) Cycling stability of LFP|Li cell with ILE, IL GPE, and 1.5BN GPE at 80 °C and 0.5C.

LFP cell with 1.5BN GPE delivers a higher discharge capacity of 142.4 and 132.8 mAh g⁻¹ at 0.1C and 0.2C, respectively. Typical capacity–voltage profiles of LFP|1.5BN GPE|Li cell displays a well-defined voltage plateau (Figure S13, Supporting Information). At the 60th cycle, the cell with AFBNNS ion gel still delivers a discharge capacity of 122.4 mAh g⁻¹, representing 92.2% capacity retention with respect to the 4th cycle and an average Coulombic efficiency (CE) of 99.4%. Although the cell with IL GPE displays comparable specific capacity at initial cycles, it shows rapid capacity decay during subsequent cycling, and finally retains only 53.5% of the specific capacity of the 4th cycle and an average Coulombic efficiency of 98.4%. Moreover, the LFP|1.5BN GPE|Li cell shows good rate capability, demonstrating the viability of AFBNNS addition in the ion gel (Figure S14, Supporting Information). The 1.5BN GPE was also applied in the cell with different cathode materials possessing higher cutoff voltage. Figure 6b,c shows cycling performances of LiNi_{1/3}Co_{1/3}Mn_{1/3}O₂ and LiCoO₂ cathode in the voltage window of 3.0–4.3 V at 0.1C, respectively. Both the NCM|Li cell and the LCO|Li cell with 1.5BN GPE outperform the cells with ILE and IL GPE, delivering discharge capacities of 134.6 and 117 mAh g⁻¹ at the 2nd cycle respectively. The NCM|Li cell and the LCO|Li cell with 1.5BN GPE also exhibit good charge–discharge voltage profiles (Figure S13, Supporting Information). The EIS analysis was conducted in the cells before cycling, and after 30 cycles to explore the stability of interfacial transporting behavior, where LFP|Li cell using ILE, IL GPE, and 1.5BN GPE are tested (Figure 6d–f). The pristine LFP|IL GPE|Li cell shows the highest interfacial resistance related to charge transfer (R_{ct}) at around 1440 Ω, while the cells with ILE and 1.5BN GPE display considerably lower interfacial resistance of 408 and 552 Ω, respectively. After cycling 30 times, the R_{ct} of the cells with ILE significantly decrease to 88 Ω (78.5% decrease) and the R_{ct} of the cells with IL GPE also show a decrease of 11%. The strong decrease of the R_{ct} can be explained by a large increase in the surface area of the lithium anode during cycling, indicating the growth of the Li dendrites.^[31] However, no obvious decrement

appears in the EIS spectra of the cells with 1.5BN GPE before and after cycling. This result clearly demonstrates the advantages of the BN GPE on promoting a uniform Li deposition. The performance of 1.5BN GPE under high temperature was also investigated. LFP|Li cells using ILE, IL GPE, or 1.5BN GPE were further tested at elevated temperature of 80 °C and 0.5C (Figure 6g). Notably, LFP|1.5BN GPE|Li cell can deliver high discharge capacity of 162.1 mAh g⁻¹ at 2nd cycle, and retain 91.3% of specific capacity after the 50th cycle, while the cells with IL GPE possess a discharge capacity of 136.6 mAh g⁻¹ and a capacity retention of 64.1% after the 50th cycle. The inferior capacity and CE fluctuation of the IL GPE cell might be attributed to the loss of solid property at the high temperature (Figure 1e). Therefore, battery performances of LFP, NCM, and LCO cell confirm the viability of AFBNNS added ion gel as an efficient solid-state electrolyte for high-performance batteries.

3. Conclusion

Robust ion gel electrolytes with improved Li-ion transport have been prepared by the addition of AFBNNS. Due to the strong interaction between the functionalized BN nanosheets and TFSI⁻ counter anions, Li-ion mobility in the ion gel is significantly enhanced. Moreover, the BN nanosheets also reinforce the mechanical strength and thermal stability of ion gels, which would be beneficial for the safety and cycle life of lithium metal batteries. The optimized 1.5BN GPE shows enhanced ionic conductivity of 6.47×10^{-4} S cm⁻¹ with a higher t_{Li^+} of 0.23 at 25 °C and a stable Li deposition behavior. An LFP|Li cell employing 1.5BN GPE delivers a high discharge capacity of 132.8 mAh g⁻¹ at 0.2C as well as a good performance with NCM and LCO cathodes. The viscosity of the ILE is responsible for the unsatisfactory rate performance. Hence, for the safer and high energy density batteries, future research will be motivated to improve the rate capability of ILE at room temperature, without compromising the unique advantages of ILs.

4. Experimental Section

Ion Gel Preparation: The AFBNNS added ion gel electrolyte was prepared by in situ thermal polymerization with ILE and AFBNNS. ILE was first prepared by mixing *N*-propyl-*N*-methylpyrrolidinium bis(trifluoromethanesulfonyl)imide (PYR₁₃ TFSI, Solvionic, 99.5%) IL and lithium bis(trifluoromethanesulfonyl)imide (LiTFSI, Aldrich, 99%) lithium salt (1 mol L⁻¹) and stirred vigorously at 45 °C over 24 h, which was used as liquid medium of the ion gel. AFBNNS and BN nanoparticles were prepared by the controlled mechanochemical synthesis process.^[32] The ion gel electrolytes without AFBNNS were fabricated via thermal polymerization. PVDF-HFP (Aldrich, *M_w* ≈ 400000) was dissolved in acetone with a concentration of 10 wt% and stirred overnight. ILE and prepared PVDF-HFP solution, TMPTA (Aldrich) were mixed in a 20 mL vial as the precursor. The weight ratio of ILE, PVDF-HFP, and TMPTA was 7:2:1. Azobisisobutyronitrile (AIBN) in toluene solution (Aldrich) was used as the thermal initiator, where the amount of AIBN was fixed to 1 wt% based on TMPTA. The prepared precursor solution was mixed with bath sonication for 10 min and stirred vigorously for 1 h. The mixed solution was then casted onto the glass plate and heated at 65 °C for 24 h. A transparent and flexible thin film was obtained and denoted as IL GPE. The AFBNNS added ion gel was also prepared by mixing BN additives with varying addition amounts from 0.5 to 2.5 wt% into precursor solution using the same procedure. Ion gel with BN nanosheet addition would be expressed as xBN GPE (*x* = 0.5–2.5). The thickness of the as-prepared ion gel was around 205 ± 5 μm and was used for all the other characterization. All the procedures were carried out in an Ar-filled glovebox.

Characterizations: Microstructure and cross-sectional images of ion gel were characterized using a field-emission scanning electron microscope (FE-SEM, SUPRA 55VP, ZEISS). The crystalline nature of the prepared ion gel was investigated with XRD spectroscopy (X'pert Powder, Panalytical) and DSC (TA Q200, TA instruments). For the DSC measurement, the ion gel sample was heated from –80 to 200 °C with a heating rate of 10 °C min⁻¹ under the nitrogen atmosphere. Thermal stability of ion gel was evaluated by TGA (TA Q50, TA instruments) from room temperature to 600 °C at a heating rate of 10 °C min⁻¹ under a nitrogen atmosphere. The polymerization of ion gel and interaction between components were confirmed using FT-IR (Vertex 70, Bruker) with attenuated total reflectance (ATR) mode. Raman spectroscopy (inVia Raman microscope, Renishaw) was employed to investigate interaction in the prepared ion gel, where a 633 nm laser source was used for excitation. Solid-state ⁷Li and ¹⁹F NMR (Avance III 300WB, Bruker) spectra were collected to investigate Li⁺ and TFSI⁻ ion mobility, and the *T*₁ relaxation times were measured using a saturation recovery pulse sequence. Mechanical stability of the prepared ion gel was evaluated with UTM (Instron 5967, Instron) and DMA (TA Q800, TA instruments). For the UTM measurement, ion gel samples of gauge length of 2.5 cm and a width of 1 cm were tested at a tensile stretch rate of 6 mm min⁻¹. For the DMA measurement, stress was applied at a constant frequency of 1 Hz. Data were collected from 30 to 80 °C at a heating rate of 5 °C min⁻¹ in air. Depth versus load experiment was also performed using nanoindenter (Hysitron TI 950 TriboIndenter, Bruker) with a diamond Berkovich tip to evaluate the surface hardness. A loading force of 100 μN was applied to two different areas of 40 discrete points, where the distance between each point was 50 μm. Average hardness and reduced modulus were obtained from the results of 80 points. Electrochemical stability of ion gel was investigated by linear sweep voltammetry (LSV) measurements with the scan rate 1 mV s⁻¹ in the potential range from 0 to 5.5 V. The ionic conductivity was measured using EIS in the frequency range of 1–10⁶ Hz and temperature range from 25 to 90 °C, with the AC amplitude value of 10 mV. EIS and the LSV were conducted using an electrochemical workstation (Solartron 1260, Solartron). Symmetric cells were prepared by sandwiching the ion gel electrolyte between two stainless steel electrodes. CR2025-type coin cells were used for EIS measurement. The stainless steel working electrode with Li counter and reference electrode in the CR2025-type coin cell were used for LSV measurement. The ionic conductivity was calculated according to the following equation

$$\sigma = L/(R \cdot A) \quad (2)$$

where σ is the ionic conductivity (mS cm⁻¹), L is the thickness of ion gel electrolyte (cm), R is the resistance (Ω) obtained by intercept of *x*-axis in the Nyquist plot, and A is the active area of ionogel electrolyte (cm²). Li⁺ transference number (*t*_{Li⁺}) was estimated by combination of DC polarization and AC impedance measurement in the Li symmetric cell.^[33] The *t*_{Li⁺} was calculated by the following equation

$$t_{Li^+} = (I_s(\Delta V - I_o R_o)) / (I_o(\Delta V - I_s R_s)) \quad (3)$$

where ΔV is the applied polarization voltage (10 mV), I_o is the initial current, and I_s is the steady state current during polarization measurement. R_o and R_s are the charge transfer resistance of cell before and after polarization, respectively. The Li plating behavior with current density distribution was simulated using COMSOL Multiphysics 5.5 software, and the parameters and calculation methods are further explained in Figure S15 and Table S1 in the Supporting Information.

Battery Cell Characterizations: Li deposition and stripping behavior of the electrolytes were investigated using galvanostatic cycling of CR2025-type Li symmetric cells. The cathode was prepared using a typical slurry casting method. A mixture of active material LFP (MTI), carbon black, and PVDF in *N*-methyl-2-pyrrolidone (NMP) were grounded in an agate mortar to obtain a uniform slurry. The weight ratio of LFP, carbon black, and PVDF binder was 75:15:10. The as-prepared slurry was then casted on carbon-coated Al foil by doctor blading and dried at 80 °C for 24 h. NCM (MTI) and LCO (MTI) cathodes were also prepared by the same preparation method. The mass loading of active material was 2.1–2.6 mg cm⁻². The prepared cathodes with Li chip as anode were assembled into CR2025-type coin cells for battery performance evaluation. To ensure interfacial contact between the cathode and ion gel electrolyte, about 8 mg of ILE was dropped onto the electrode and excess of ILE was carefully removed by the dry absorbent. For the assembly of the ILE-based cell, Celgard2400 was used as a separator. Galvanostatic cycling at various rates was conducted with the battery testing system (CT2001A, LANHE). All the coin cells were assembled in the Ar-filled glovebox.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

boron nitride, ionic liquids, lithium-ion transference number, lithium metal batteries, solid-state electrolytes

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